Feasibility Study for

LeitDrive 690V new IGBT module

Evaluation of FF1200R17IP5P with DB1.00 and DB1.51

Outline

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Motivation

Original Design Target (Wind)

- $I_{\text{nom}}^{\text{inv}} = 370\,\text{A}$
- $f_{\text{pwm}}^{\text{inv}} = 2.5\,\text{kHz}$
- Overload $\equiv$ wind gust $\Rightarrow I_{\text{max}}^{\text{inv}} \approx I_{\text{nom}}^{\text{inv}}$

New Requirements (Drive Applications)

- $I_{\text{max}}^{\text{inv}} = 1.4 \times I_{\text{nom}}^{\text{inv}} = 518\,\text{A}$
- $f_{\text{pwm}}^{\text{inv}} = 4\,\text{kHz}$ (hard parallelization)

Design Constraint

$$T_{\text{j}}^{\text{design}}(t)\Big|_{\text{max}} \leq T_{\text{j}}^{\text{max}} - 50\,^{\circ}\text{C}$$

Conclusion

Existing modules (DP650B1700, FF650R17IE4) are no longer adequate.

Candidate replacement: FF1200R17IP5P (Infineon, 1200 A / 1700 V, $T_{\text{j}}^{\text{max}} = 175\,^{\circ}\text{C}$).

Modules Under Test

Module	**Manufacturer**	I_{nom}	$T_{\text{j}}^{\text{max}}$	**Status**
DP650B1700	Danfoss	650 A	150 °C	Series production
FF650R17IE4	Infineon	650 A	150 °C	Series production
FF650R17IE4DB2	Infineon	650 A	150 °C	Series production
FF1200R17IP5P	**Infineon**	1200 A	175 °C	**Candidate**

Driver Boards

- **DB1.00** – series production
- **DB1.51** – candidate (new)

Gate Resistances

	R_{g}^{on}	$R_{\text{g}}^{\text{off}}$
DB1.00	1.24 Ω	3.6 Ω
DB1.51	1.175 Ω	1.175 Ω

DPT Configurations

Test	IGBT Module	Driver Board	Type
DPT 1	DP650B1700	DB1.00	Baseline (production)
DPT 2	DP650B1700	DB1.51	New driver board
DPT 3	FF1200R17IP5P	DB1.00	New module
DPT 4	FF1200R17IP5P	DB1.51	**Full candidate config.**

DPT Conditions

$$I_c = 620\,\text{A}, \quad V_{ce} = 630\,\text{V}, \quad T_{case} = 25\,^\circ\text{C}$$

Power Loss Analysis – Operating Conditions

Scenario	$I_{\text{load}}^{\text{inv}}$	$f_{\text{pwm}}^{\text{inv}}$	$f_{\text{load}}^{\text{inv}}$	$T_{\text{hs}}^{\text{inv}}$	$\cos(\varphi)$	**DB**
1	370 A (100%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.00
2	520 A (140%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.00
3	370 A (100%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.51
4	520 A (140%)	4 kHz	15 Hz	65 °C	+0.8 (drive)	DB1.51
5	370 A (100%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.00
6	520 A (140%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.00
7	370 A (100%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.51
8	520 A (140%)	4 kHz	15 Hz	65 °C	−0.8 (gen.)	DB1.51

Junction Temperature – Drive Mode ($\cos\varphi = +0.8$), DB1.00

Scenario 1 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.00

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	120 °C	97 °C	242 W	54 W	576 W	175 W	25 °C
FF650R17IE4DB2	130 °C	95 °C	245 W	50 W	598 W	196 W	32 °C
FF1200R17IP5P	110 °C	83 °C	172 W	42 W	705 W	170 W	18 °C

Scenario 2 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.00

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	156 °C	108 °C	420 W	85 W	911 W	220 W	42 °C
FF650R17IE4DB2	170 °C	103 °C	430 W	77 W	950 W	236 W	50 °C
FF1200R17IP5P	131 °C	89 °C	280 W	64 W	1000 W	220 W	27 °C

Junction Temperature – Drive Mode ($\cos\varphi = +0.8$), DB1.51

Scenario 3 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.51

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	121 °C	98 °C	242 W	54 W	572 W	175 W	25 °C
FF650R17IE4DB2	128 °C	95 °C	245 W	49 W	585 W	198 W	30 °C
FF1200R17IP5P	107 °C	84 °C	172 W	42 W	644 W	171 W	18 °C

Scenario 4 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.51

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	156 °C	110 °C	421 W	86 W	905 W	220 W	45 °C
FF650R17IE4DB2	169 °C	103 °C	431 W	77 W	927 W	240 W	50 °C
FF1200R17IP5P	127 °C	90 °C	280 W	65 W	913 W	222 W	25 °C

Junction Temperature – Generator Mode ($\cos\varphi = -0.8$), DB1.00

Scenario 5 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.00 *(worst case, wind)*

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	107 °C	124 °C	62 W	210 W	555 W	220 W	30 °C
FF650R17IE4DB2	112 °C	114 °C	62 W	189 W	568 W	215 W	25 °C
FF1200R17IP5P	102 °C	94 °C	45 W	159 W	688 W	179 W	13 °C

Scenario 6 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.00 *(worst case, wind)*

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	130 °C	155 °C	100 W	343 W	841 W	283 W	50 °C
FF650R17IE4DB2	137 °C	134 °C	104 W	299 W	862 W	277 W	35 °C
FF1200R17IP5P	118 °C	107 °C	72 W	247 W	964 W	240 W	32 °C

Junction Temperature – Generator Mode ($\cos\varphi = -0.8$), DB1.51

Scenario 7 – $I_{\text{load}}^{\text{inv}} = 370\,\text{A}$, DB1.51 *(worst case, wind)*

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	106 °C	124 °C	62 W	210 W	548 W	200 W	22.5 °C
FF650R17IE4DB2	111 °C	114 °C	62 W	189 W	557 W	219 W	22.5 °C
FF1200R17IP5P	100 °C	94 °C	45 W	159 W	629 W	181 W	12 °C

Scenario 8 – $I_{\text{load}}^{\text{inv}} = 520\,\text{A}$, DB1.51

	T_{j}^{Q}	T_{j}^{D}	P_{c}^{Q}	P_{c}^{D}	P_{sw}^{Q}	P_{sw}^{D}	ΔT_{j}
FF650R17IE4	129 °C	155 °C	103 W	343 W	836 W	283 W	50 °C
FF650R17IE4DB2	136 °C	135 °C	104 W	299 W	843 W	282 W	35 °C
FF1200R17IP5P	114 °C	107 °C	72 W	247 W	878 W	243 W	30 °C

Switching Energy – Reference vs. Measured

Reference Values (from datasheet, scaled to test conditions)

	T_{case}	E_{on}^{ref}	E_{off}^{ref}	I_c	V_{ce}
FF650R17IE4	25 °C	136 mJ	93 mJ	620 A	630 V
FF1200R17IP5P	25 °C	132 mJ	100 mJ	620 A	630 V

Measured – DB1.00

	E_{on}^{m}	E_{off}^{m}
FF650R17IE4	80 mJ	90 mJ
FF1200R17IP5P	105 mJ	152 mJ

Measured – DB1.51

	E_{on}^{m}	E_{off}^{m}
FF650R17IE4	91 mJ	93 mJ
FF1200R17IP5P	107 mJ	130 mJ

DPT Summary – Switching Parameters

	Case 1	Case 2	Case 3	Case 4
	DP650 + DB1.00	DP650 + DB1.51	FF1200 + DB1.00	FF1200 + DB1.51
E_{or}	80	91	105	107
E_{off} [mJ]	90	93	152	**130**
I_{rr} |	512	**448**	488	**464**
Δu_{ce} [V]	245	187	173	**86**

FF1200 + DB1.00 (Case 3)

Higher E_{off} (152 mJ) compared to reference.
Asymmetric R_g may be sub-optimal.

FF1200 + DB1.51 (Case 4)

Best Δu_{ce} (86 V). Reduced E_{off} vs Case 3.
Lowest voltage overshoot overall.

Results Overview – FF1200R17IP5P vs. Existing Modules

Metric	FF650R17IE4	FF650R17IE4DB2	FF1200R17IP5P
T_j^{max} allowed	150 °C	150 °C	175 °C
T_j^{Q} @ 520 A, drive	156 °C	170 °C	131 °C
T_j^{D} @ 520 A, gen.	155 °C	134 °C	107 °C
ΔT_j @ 520 A, drive	42 °C	50 °C	27 °C
Design rule satisfied (140% load)	**No**	**No**	**Yes**

Conclusions

FF1200R17IP5P + DB1.51 – Recommended

- Satisfies the thermal design rule at 140% load
- Lower T_j than existing modules in all scenarios
- Lowest $\Delta u_{ce} = 86\,\mathrm{V}$ (Case 4)
- Compatible with DB1.51 driver board
- $T_j^{max} = 175\,^\circ\mathrm{C}$ provides larger thermal margin

Existing Modules – Inadequate

- FF650R17IE4 and FF650R17IE4DB2 exceed the thermal limit at 140% load
- $T_j^{max} = 150\,^\circ\mathrm{C}$ insufficient for new requirements

Next Steps

- Load testing under real operating conditions
- Thermal cycling validation
- Platform integration testing

Thank you for your attention

Questions?

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690 V LeitDrive Platform – IGBT Module Feasibility Study

FF1200R17IP5P (Infineon) – DPT Evaluation with DB1.00 and DB1.51